

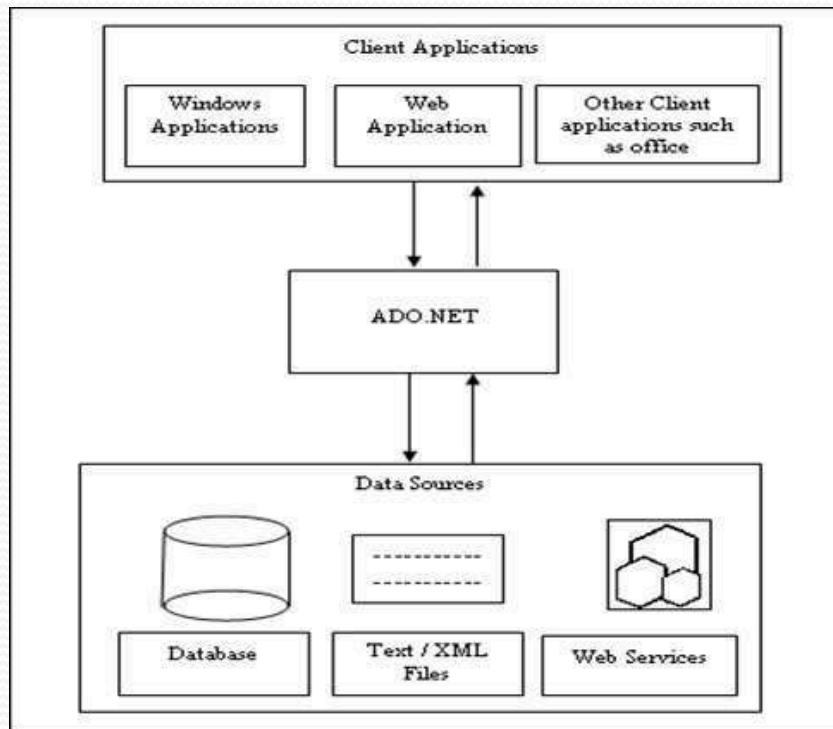
Unit	PARTICULAR	Marks
IV	ADO.Net Architecture Of ADO.Net, Data Base Manipulation, .Net Data Provider, Data Adapter, Data Set, Data Table, Introduction about SQL server. Connection with SQL server. Command, Data Reader, Data Grid View, Execute reader, Execute Non Query, Execute Scalar. Crystal Reports Syllabus: Introduction to Crystal Reports and its integration with Visual C#, Designing reports and working with data sources, Formatting, grouping, Adding parameters, and summarizing report data,Exporting and deploying reports in different formats.	25%

What is ADO.net?

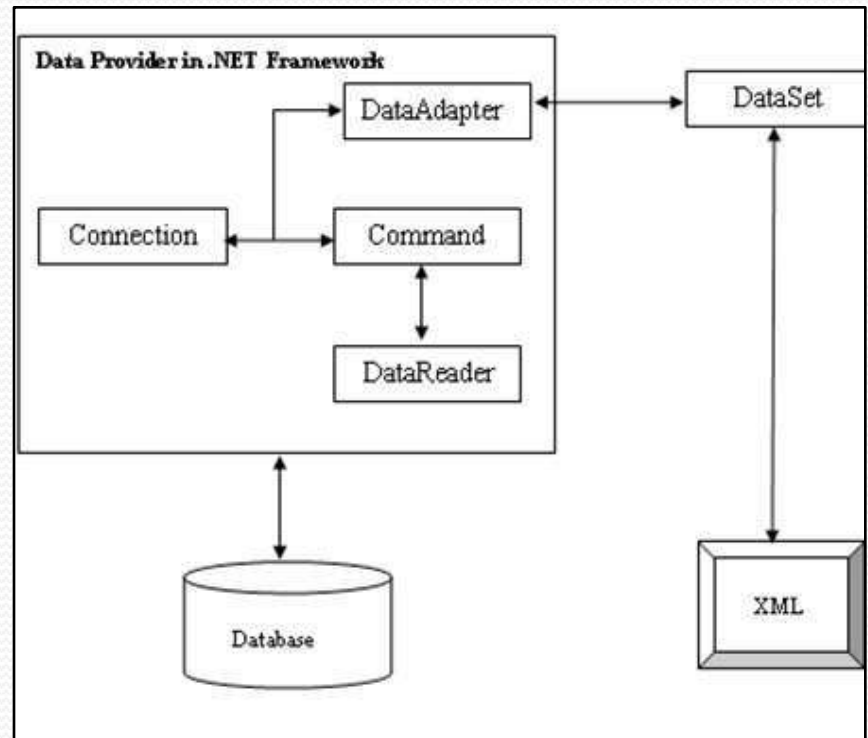
- *The full-form of ADO.net is ActiveX Data Object.*
- *ADO.net is the data-access technology that enables applications to connect to data stores and manipulate data contained in them in various ways.*
- *It is based on the .NET framework and it is highly integrated with the rest of the framework class library.*
- *ADO.net is a collection of classes, interfaces, structures, enumerated types that manage data access from the relational data stores within the .NET framework.*

ADO.NET

Understanding ADO.net



The ADO.NET Object Model



WorkingwithAdo.net

→ Many of web application development, in many of the developers data situations involves opening data store, making request for specific data, and then populating the table in the browser with the data.

→ To address the above situations ADO.NET supports the following classes:

1) *DataSet*

2) *DataReader*

Working with ADO.NET namespaces

→ The six core ADO.NET namespaces are;

- 1) System.Data
- 2) System.Data.Common
- 3) System.Data.OleDb
- 4) System.Data.Odbc
- 5) System.Data.SqlClient
- 6) System.Data.SqlTypes

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- The classes from the above namespaces are categorized as follow:
 - Disconnected: These Class provides basic structure of ADO.NET includes DataTable, etc. The object of this class are capable of storing data without any dependency on specific data provider.
 - Shared: Set of base classes for data provider and shared amount all data provider. Includes DataSet, DataRow, DataTables, Constraint, etc.
 - Data Provider : These classes are meant to work with different data sources. Performs all database management operations as specific database.
 - E.g.: SqlConnection data provider works only with SQL Server Database.

Type of Architecture

→ *ADO.NET is both Connection oriented as well as Disconnection oriented*

* *Connected: Forward-only, Read-only, mode.*

*Application issues query then reads back results and processes them.
DataReader object.*

* *Disconnected:*

*Application issues query then retrieve and stores results for processing
Minimizes time connected to database DataSet object.*

→ *DataProvider:-(ConnectedEnvironment)*

Adataproviderisusedforconnectingtoadatabase,retrievingdata,storingthe data in a dataset, reading the retrieved data, and updating the database.

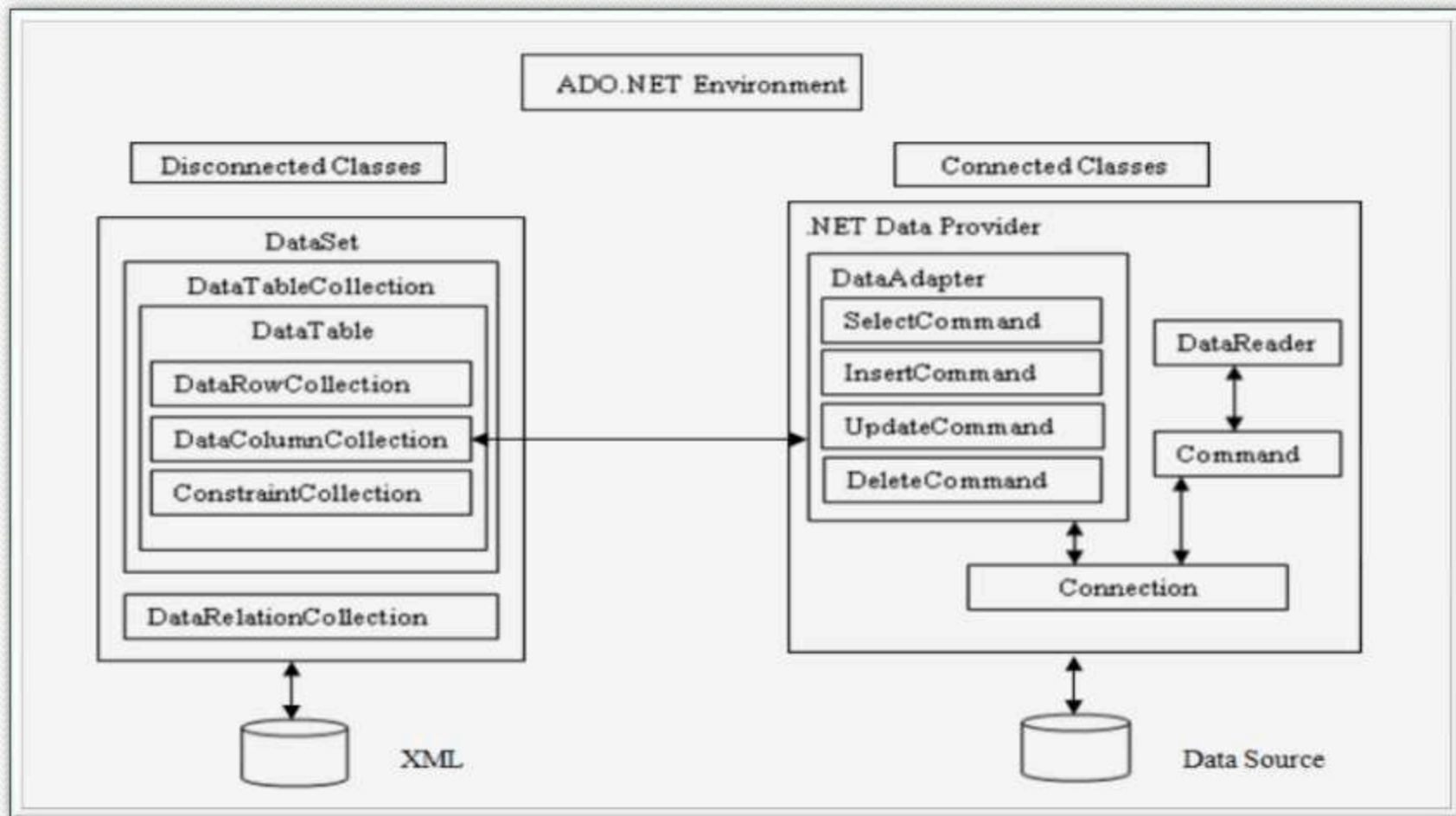
- * Connection:-Thiscomponentisusedtoestablishaconnectionwitha datasourcesuchasdatabase.*
- * Command:-Thiscomponentisusedtoretrieve,insert,delete,ormodify data in a data source.*
- * DataReader:-Thiscomponentisusedtoretrievedatafromadata source in a read-only and forward-only mode.*
- * DataAdapter:-Thiscomponentisusedtotransferdatatoandfrom database.Adatareaderretrievedatafromadatabaseintoadataset.*

→ *DataSet:-(DisconnectedEnvironment)*

Thedatasetisamemory-basedrelationalrepresentationofdata.

- * DataTableCollection:-Itcontainsalltable retrievedfromthedata source.*
- * DataRelationCollection:-Itcontainsrelationshipandlinksbetween table in a dataset.*
- * DataTable:-Itrepresentatableinthedatatablecollectionofadataset.*
- * DataRowCollection:-Itcontainsalltherowinadatatable.*
- * DataColumnCollection:-Itcontainsallthecolumninadatatable.*

ADO.NET Environment



DataReader

→ *Adatareader(SqlDataReader,etc.,)is a simple and fastest way of selecting some data from a data source but also least capable.*

** You can't directly instantiate a data reader object. An instance is returned from appropriate Command object after having call ExecuteReader() method.*

** This data reader is forward-only , read-only connected cursor. You can only travel the records in one direction. Database connection used is kept open until the data reader has been closed.*

** The DataReader object requires a live connection to the database. So when you are done with DataReader object, be sure to call close(). If not connection stays alive.*

** When DataReader object is closed (via an explicit call to close) or the object being garbage collected, the underlying connection(con) may also be closed depending on which of the ExecuteReader() method is called.*

** If you call ExecuteReader() and pass CommandBehaviour.*

CloseConnection you can force the connection to be closed when the

reader is closed. 10

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- For *DataReader* object you have the following methods:
 - * *Read()*: Reads record by record in the data reader.
 - * *Close()*: Closes the *SqlDataReader* connection
 - * *Dispose()*: Disposes the data reader object.
 - Some of the important properties of *DataReader* object are:
 - * *Connection*: Gets the *SqlConnection* associated with the *SqlDataReader*.
 - * *HasRows*: Gets the value that indicates whether the *SqlDataReader* contains one or more rows.
 - * *IsClosed*: Retrieves a boolean value that indicates whether the specified *SqlDataReader* instance has been closed.

DataAdapter

- *The DataAdapter classes (Eg: SqlDataAdapter,..) bridge the gap between the disconnected DataTable objects and the physical data source.*
- *The SqlDataAdapter provides two way data transfer mechanisms:*
 - * It is capable of executing SELECT statement on a data source and transferring the result set into a DataTable object.*
 - * It is also capable of executing the standard INSERT, UPDATE and DELETE statements and extracting the input data from DataTable object.*



→ The commonly used properties of `SqlDataAdapter` class are:

- * `Select Command`: Gets or sets an object of type `SqlCommand`. This command is automatically executed to fill a `DataTable` with the result set.
- * `Insert Command`: Gets or sets an object of type `SqlCommand`.
- * `Update Command`: Gets or sets an object of type `SqlCommand`.
- * `Delete Command`: Gets or sets an object of type `SqlCommand`.

→ This `SqlDataAdapter` class also provides a method called `Fill()`.

- * Calling the `Fill()` method automatically executes the command provided by the `Select Command` property and retrieves the result set and copies it to `DataTable`.

Using Parameter

→ If you requires SQL Statements that required you to configure using parameter then you need to instantiate `Parameter` (i.e., `SqlParameter`, ...) object and providing it with necessary information such as name, value, type, size, direction, etc.,

→ The following are properties of the `SqlParameter` class:-

* `ParameterName`, `SqlDbType`, `Size`, `Direction`, `SourceColumn`, `value`

→ Coding Snipped in using parameters are:

```
for e.g. : SqlCommand myCmd = new SqlCommand();  
myCmd.CommandText="SELECT * FROM CUSTOMERS WHERE  
CITY=@CITY";  
SqlParameter parmCity = new SqlParameter();  
parmCity.ParameterName="@CITY";  
parmCity.SqlDbType=SqlDbType.varchar;  
parmCity.Direction=ParameterDirection.input;  
parmCity.Value="Ahmedabad"  
myCmd.Parameters.Add(parmCity);
```

DataSet

→ The *DataSet* class has been designed as an offline container of data.

* Data held within the *DataSet* is not necessary to come from database. It can be recorded from database. It can be recorded from CSV file, XML source, etc.

* A *DataSet* class contains a set of *DataTable* classes and links between tables.

fore.g.:

```
foreach(DataRow row in ds.Tables["Customers"].rows);  
    Console.WriteLine("{0} and {1}", row[0], row[1]);
```

→ Each *DataTable* class consists *DataColumn*, *DataRow*, *Constraints* and *ExtendedProperties* classes.

→ .csv = Comma Separated Values

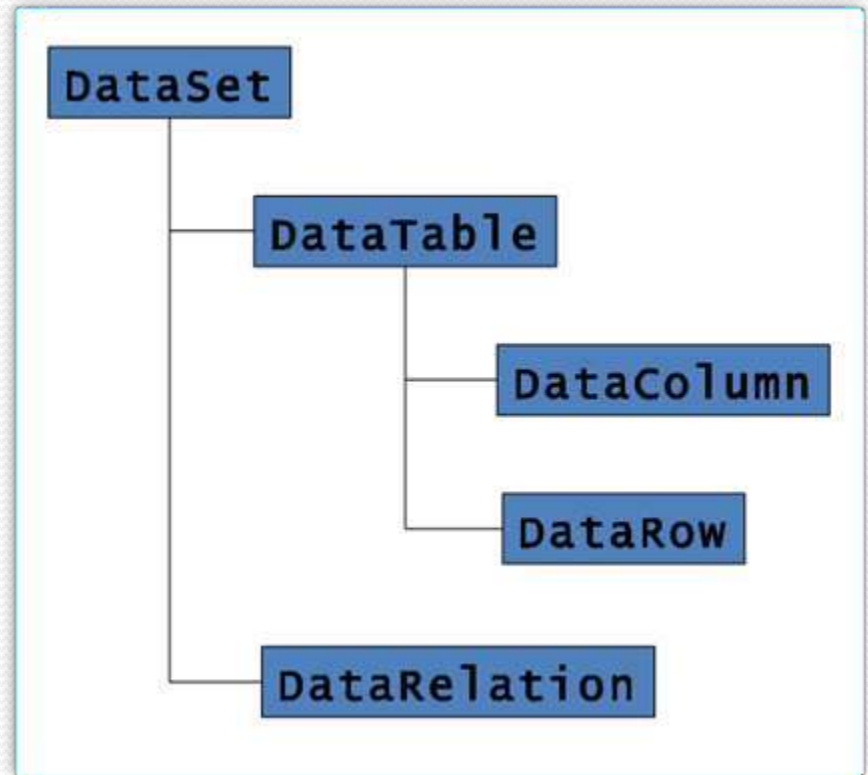
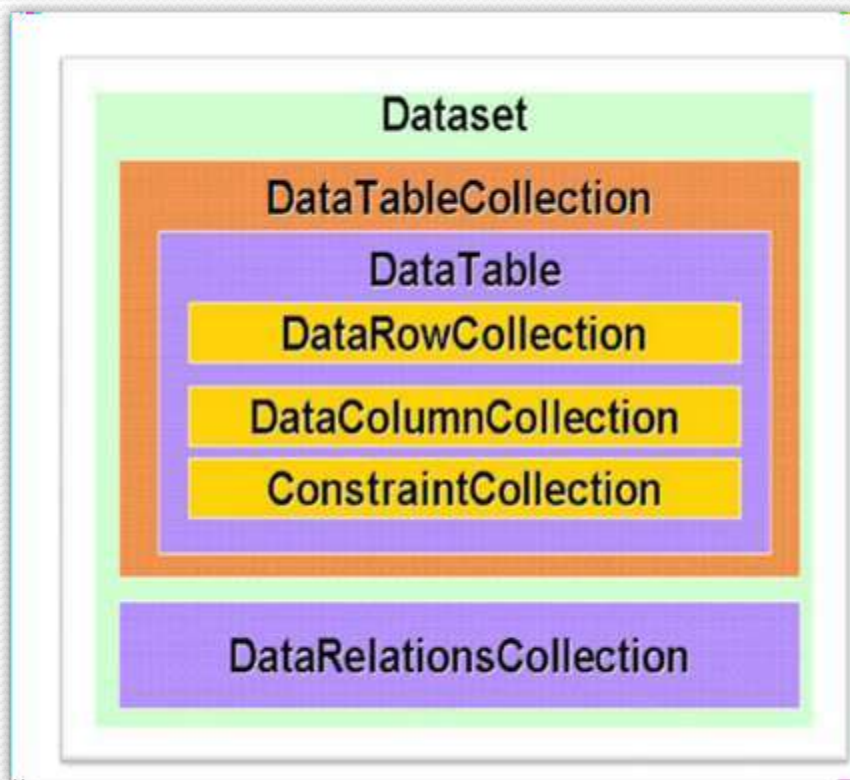
→ Here DataColumn object defines the properties of a column within DataTable. These properties include datatype(Int32, Char, DateTime, etc.), read-only. You can also assign a name for column.

fore.g.:

```
DataColumn customerID=new DataColumn("CustomerID",typeof(int));  
customerID.AllowDBNull=false;  
customerID.ReadOnly=false;  
customerID.AutoIncrement=true;  
customerID.Unique=true;
```

→ The actual data within a table is accessed via a DataRow object. One of the interesting aspect of DataRow is that it is versioned. This permits you to receive various values for a given column in a particular row.

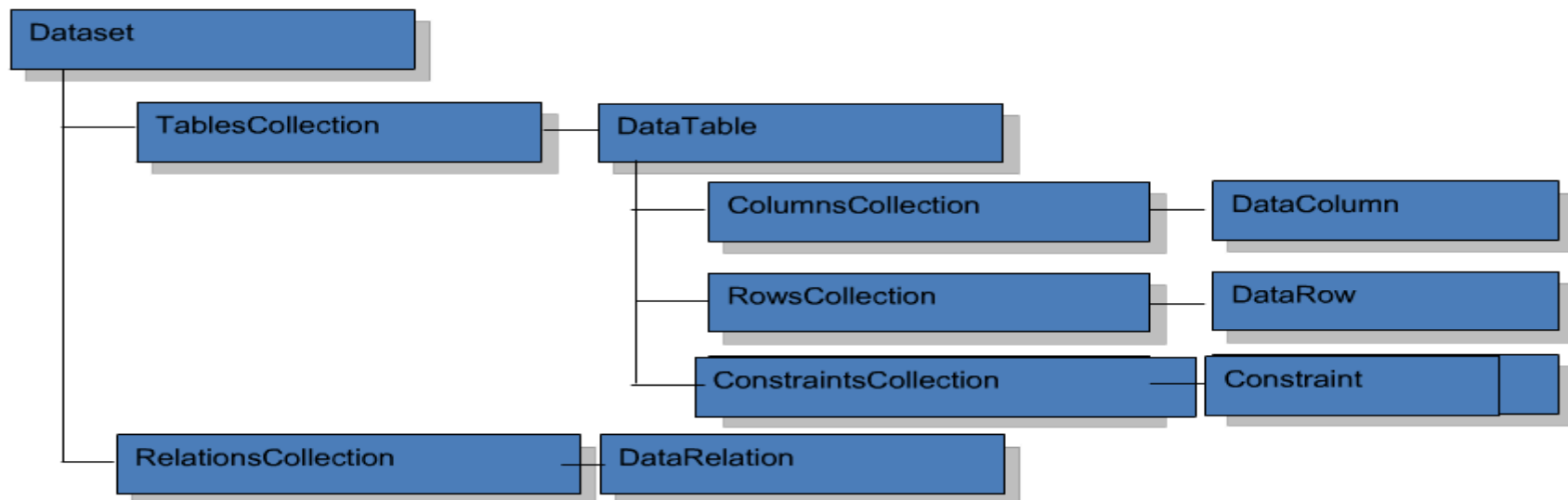
DataSet



DataSet Architecture

→ You can create DataSet in any one of the following methods:

- * Using Server Explorer
- * Using DataAdapter
- * Programmatically
- * Using Xml



→ CodingsnippetofusingDataSet:

for e.g. :

```
SqlConnection con = new SqlConnection(...);  
SqlCommand cmd = new SqlCommand(...);  
SqlDataAdapter da = new SqlDataAdapter(cmd, con);  
DataSet ds = new DataSet();  
da.Fill(ds, "MyTable");
```

→ BetterguidelineregardingDataSetis:

- * Don't useDataSetobjectunlessyourprojectrequiredtoworkwith multipleDataTableobjectandyouneedacontainerforthoseobject.
- * IfyouareworkingwithsingletablethenjustuseDataTableobject rather than DataSet.

→ YoucancreateschemasforaDataTableinthreeways.

- * Lettheruntimedoitforyou.
- * Writecodetocreatetables.
- * Use theXMLschemagenerator.

→ The *DataAdapter* class is used to place the data in a *DataSet* class. This Adapter class issues the SQL clause and fills a table within the *DataSet*.

**fore.g.:*

```
SqlDataAdapter da = new SqlDataAdapter(strCmd, con);
```

```
DataSet ds = new DataSet();
```

```
da.Fill(ds, "Customers"); // Populating DataSet with DataAdapter.
```

→ The *DataSet* class has been designed to establish relationships between data tables with ease.

→ At present you can apply the *ForeignKeyConstraint* and *UniqueConstraint* on a column in a *DataSet*.

→ You can also set *Update* & *Delete* constraint on a column that have constraint relationships.

→ *These rules are:*

**Cascade, None, SetDefault, SetNull.*

fore.g.:

```
DataTable dt = GetDataTable(); // It returns DataTable  
DataColumn[] pk = new DataColumn[1]; pk[0]  
= dt.Columns["StudentID"]; dt.PrimaryKey =  
pk;
```

→ *The constraints are enforced within a DataSet class if the
EnforcedConstraints property of the DataSet is true.*

DataProvidersClasses

→ In .NET framework data provide is used for connecting to database, executing commands and retrieving results. By default the following data provides are included in .NET framework :

- * *DataProviderforSQLServer*
- * *DataProviderforOLEDB*
- * *DataProviderforODBC*
- * *DataProviderforOracle*

→ In .NET to access the dataset of classes defined in the namespaces *System.Data* and other namespaces defined within this namespaces.

→ Each data provider contains the following objects:

- * *Connection* (fore.g.: *SqlConnection*, *OracleConnection*, *OdbcConnection*, etc.)
- * *Command*, *CommandBuilder* (fore.g.: *SqlConnection*, *SqlConnectionBuilder*, etc.)
- * *DataAdapter* (fore.g.: *SqlDataAdapter*, etc.)
- * *DataReader* (fore.g.: *SqlDataReader*, etc.)
- * *Parameter* (fore.g.: *SqlParameter*, etc.)
- * *Transaction* (e.g.: *SqlTransaction*, etc.)

→ Each Provider uses different namespaces. Their namespaces are in respective :

- * *System.Data.SqlClient*
- * *System.Data.OleDb*
- * *System.Data.Odbc*
- * *System.Data.OracleClient*

→ In addition to above namespaces few more important namespaces in ADO.NET are:

- * *System.Data*
- * *System.Data.Common*
- * *System.Data.SqlTypes*

→ ADO.NET consists number of classes that are used regardless you are using the SQL Server Classes or OLEDB classes.

→ The *System.Data* namespace consists the following classes:

- * *DataSet*
- * *DataTables*
- * *DataRow*
- * *DataColumn*
- * *DataRelation*
- * *Constraint*

Classes in Data Providers

- *Each data providers support the following types of core objects to communicate with database.*
- *The Connection object provides connectivity to a data source.*
- *The Command object enable access to database commands to return data, modify data, run stored procedures, and send or retrieve parameter information.*
- *The DataReader provides a high performance stream of the data from the data source.*
- *Finally, the DataAdapter provides the bridge between the DataSet object and the data source.*
- *The DataAdapter uses Command objects to execute SQL commands at the data source to both load the DataSet with data, and reconcile changes made to the data in the DataSet back to the data source.*

ConnectionObject

- *The different Connection classes are in different data provider models.*
 - * *fore.g.: SqlConnection, OleDbConnection, OracleConnection, etc.*
- *The properties of SqlConnection class are:*
 - * *ConnectionString, DataSource, Database, State.*
 - * *In above properties DataSource, Database, State are in read-only mode.*
- *You can pass necessary connection string information to the SqlConnection in a string format or store in "web.config" file.*

→ You can store database connection parameters in configuration file under <ConnectionString> section. Here you can specify name of connection, connection string parameter and provider for this connection.

**fore.g. usage of SqlConnection:*

```
SqlConnection con = new SqlConnection("Data  
Source=FARZADFWADIA-PC;Initial Catalog=FarzadWadia;  
Integrated Security=True");
```

```
con.Open();//Use connection object.
```

```
con.Close(); //you need to call this explicitly, because .net doesn't  
implicitly release the connection when they fall out of scope.
```

CommandObject

→ In ADO.NET, a command is a string of the containing SQL statements that is to be issued to the database. It can also be a stored procedure or the name of the table which return all columns and rows.

→ A Command can be constructed by using the instance of `SqlCommand`. for e.g. :

```
SqlCommand cmd = new SqlCommand("Select * from  
Employee", con);
```

→ After building the command object you need to execute with any one of the methods:

- * `ExecuteNonQuery()`: Executes the command but doesn't return any output. Only it will return no of rows effected.
- * `ExecuteReader()`: Executes the command and returns a type `SqlDataReader`.
- * `ExecuteRow()`: Executes the command and returns `SqlRecord`, (Single row)
- * `ExecuteScalar()`: Executes the command and returns a single values.
- * `ExecuteXmlReader()`: Executes the command and returns `XmlReader`.

→ Some of the important properties of Command object are:

- * *CommandText*: Assign SQL statement or stored procedure name,
- * *CommandTimeout*: Specifies no of second to wait for executing a command
- * *CommandType*: Possible values are Text, Stored Procedure, Table direct.
- * *Connection*: Gets or Sets SqlConnection object.

→ Calling a stored procedure with a Command object for
e.g. :

```
SqlCommand cmd = new SqlCommand("MyStoredProc", con);  
cmd.CommandType = CommandType.StoredProcedure;  
cmd.Parameters.Add(new SqlParameter(...));  
cmd.Parameters[0].Value = val;  
cmd.ExecuteNonQuery();
```

ADO.NET Programming

→ ADO.NET Programming involves the following steps:

- * Create a Connection object and provide connection string information.
- * Create a Command object and provide details of SQL command that is to be executed.
- * Command can be inline SQL Text command, Stored Procedure or direct table access. Also provide required parameters if needed.
- * If your command doesn't return result set then execute the command simply by calling one of Execute methods on Command object.
- * If your command object returns a result set then you can store the result set in DataReader or DataAdapter object.
- * To load the data from DataReader to DataTable just call Load() method on DataTable.

```
SqlDataReader myReader=cmd.ExecuteReader(...);  
DataTable  
myTable=new DataTable();  
myTable.Load(myReader);
```

** In reverse if you want a reader from DataTable then use:*

DataTableReader myReader = myTable.CreateDataReader();

** If you want to retain the result set for future use without maintaining the connection to the database then use DataAdapter object. This DataAdapter object is used to fill DataSet or DataTable objects.*

** If you don't want to retain the result set, rather simply process the command in a swift fashion, you use Command object needs a live connection to database and it works as forward-only read-only cursor.*